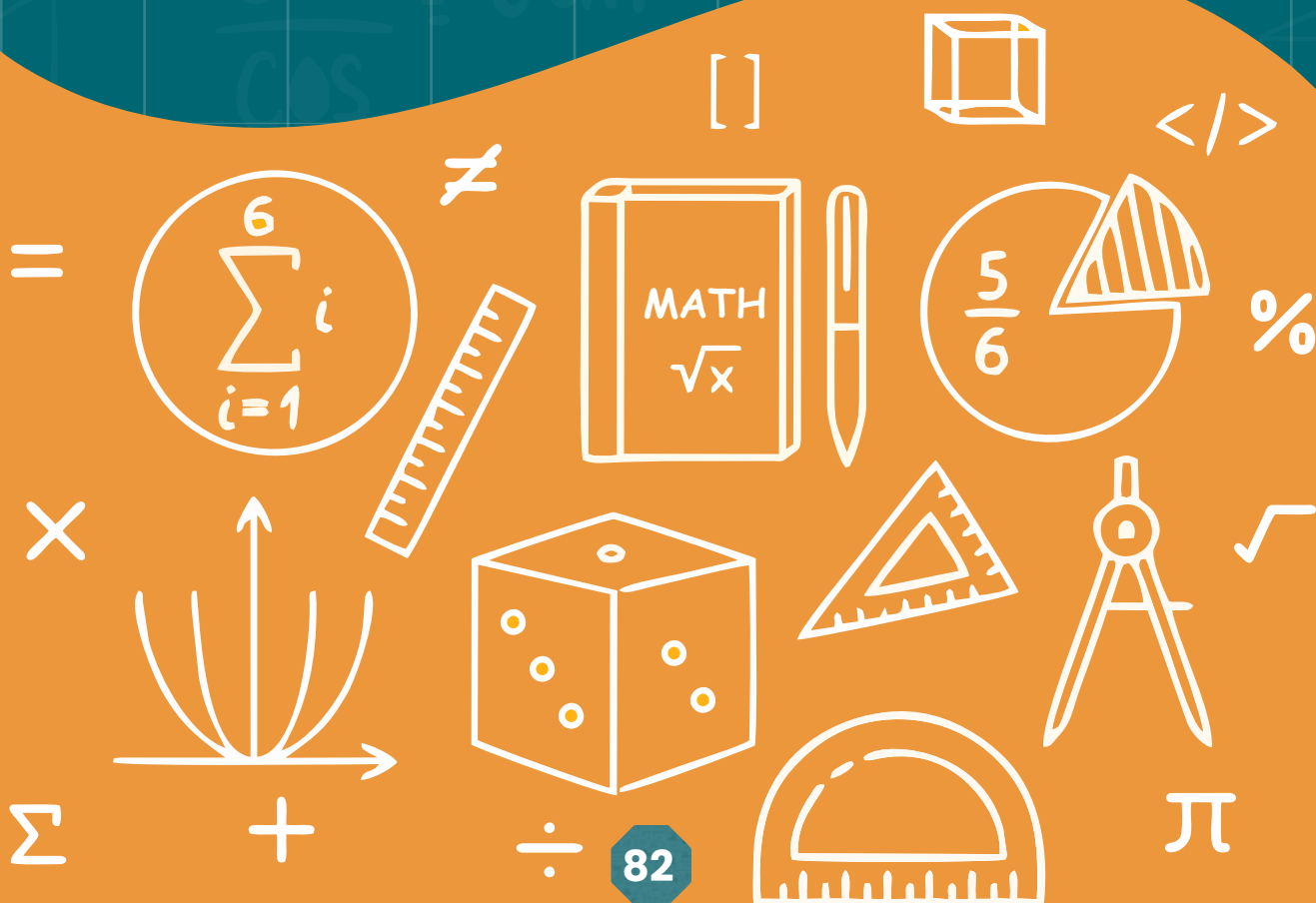


UNIT 8

ALGEBRAIC EXPRESSIONS



Evaluating Algebraic Expressions

An expression is an algebraic expression if it contains sums and/or products of variables and numbers. To evaluate an algebraic expression, replace the variable or variables with known values, and then use the order of operations. We evaluate an expression when we are given the value of a variable.

Order of Operations

Step 1: Evaluate expressions inside grouping symbols.

Step 2: Evaluate all powers.

Step 3: Do all multiplications and/or divisions from left to right.

Step 4: Do all additions and/or subtractions from left to right.

Example 1: Evaluate each expression

(i) $x - 5 + y$ if $x = 15$ and $y = -7$

$$x - 5 + y = 15 - 5 + (-7) \quad \text{Substitute.}$$

$$= 10 + (-7)$$

$$= 3$$

(ii) $6ab^2$ if $a = -3$ and $b = 3$

$$6ab^2 = 6(-3)(3^2) \quad \text{Substitute.}$$

$$= 6(-3)(9)$$

$$= (-18)(9)$$

$$= -162$$

Example 2: Evaluate if $m = -2$, $n = -4$, and $p = 5$.

(i) $\frac{2m + n}{p - 3}$

$$\frac{2m + n}{p - 3} = \frac{2(-2) + (-4)}{5 - 3} \quad \text{Substitute.}$$

$$= \frac{-4 - 4}{5 - 3}$$

$$= \frac{-8}{2} \text{ or } -4$$

(ii) $-3(m^2 + 2n)$

$$\begin{aligned} -3(m^2 + 2n) &= -3 [(-2)^2 + 2(-4)] \text{ Substitute.} \\ &= -3 [4 + (-8)] \\ &= -3 (-4) \text{ or } 12. \end{aligned}$$

Example 3: Evaluate $3(a - b) + 2(c - 2)$ if $a = 9$, $b = 4$, and $c = 3$.

$$3(a - b) + 2(c - 2) = 3(9 - 4) + 2(3 - 2) \text{ Substitute for } a, b, \text{ and } c.$$

Order of Operations:

Brackets.

Order of Operations:

Multiply.

Order of Operations:

Add.

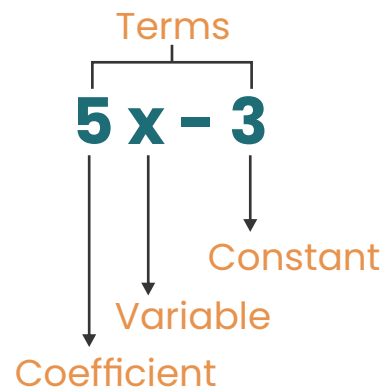
$$= 3(5) + 2(1)$$

$$= 15 + 2$$

$$= 17.$$

In the above expression (i.e. $5x - 3$),

- x is a **variable**, whose value is unknown to us which can take any value.
- 5 is known as the **coefficient** of x , as it is a constant value used with the variable term and is well defined.
- 3 is the **constant** value term that has a definite value.



TYPES OF ALGEBRAIC EXPRESSION

There are 3 main types of algebraic expressions which include:

- Monomial Expression
- Binomial Expression
- Polynomial Expression

Monomial Expression

An algebraic expression which is having only one term is known as a monomial.

Examples of monomial expressions include $3x^4$, $3xy$, $3x$, $8y$, etc.

Binomial Expression

A binomial expression is an algebraic expression which is having

two terms, which are unlike.

Examples of binomial include $5xy + 8$, $xyz + x^3$, etc.

Polynomial Expression

In general, an expression with more than one term with non-negative integral exponents of a variable is known as a polynomial.

Examples of polynomial expression include $ax + by + ca$, $x^3 + 2x + 3$, etc.

Activity: Identify and tick (✓) the given polynomial as a monomial, a binomial, or a trinomial.

S. No.	Polynomial	Monomial	Binomial	Trinomial
(i)	$3.4 + 3.4x$			
(ii)	$z^2 + 5z + 6$			
(iii)	-8			
(iv)	$2c + 5b + 6$			
(v)	$14 + x$			
(vi)	$5x - 2$			
(vii)	$4b - 2a$			
(viii)	$f^2 + 5f + 6$			

OPERATIONS WITH POLYNOMIALS

Rules for Adding Polynomials

To add polynomials, simply combine like terms or similar terms.

To combine similar terms, get the sum of the numerical coefficients and copy the same literal coefficients. For convenience, write similar terms in the same column.

Example 1: Find the sum.

(i) $5a + (2a + 6)$

Solution:

(i) $5a + (2a + 6)$

$$= (5a + 2a) + 6$$

$$= (5 + 2)a + 6$$

$$= 7a + 6$$

b. $(2x + 4) + (4x - 1)$

Group like terms.

Add the numerical coefficients of the like terms, then simplify.

(ii) $(2x + 4) + (4x - 1)$

$$= (2x + 4x) + (4 - 1)$$

$$= (2 + 4)x + (4 - 1)$$

$$= 6x + 3$$

Group like terms.

Add the numerical coefficients of the like terms, then simplify.

Example 2: Simplify.

$(5a + 8b - 2c) + (8a + 7b - 8c)$

Solution:

a. By removing the parentheses and grouping and combining like terms, we get:

$$(5a + 8b - 2c) + (8a + 7b - 8c)$$

$$= (5a + 8a) + (8b + 7b) + (-2c - 8c)$$

$$= (5 + 8)a + (8 + 7)b + (-2 - 8)c$$

$$= 13a + 15b - 10c$$

Group like terms.

Combine the numerical coefficients of each term, then simplify.

Example 3: Add $3a + 2b$ and $2a + 5b$

Solution:

Horizontal Method

$$(3a + 2b) + (2a + 5b)$$

$$= 3a + 2a + 2b + 5b$$

$$= (3 + 2)a + (2 + 5)b$$

$$= 5a + 7b$$

Vertical Method:

$$3a + 2b$$

$$2a + 5b$$

$$\hline 5a + 7b$$

Example 4: Add $6a^2b - 3ab^2$ and $4a^2b + 3ab^2$ **Solution:****Horizontal Method**

$$\begin{aligned}
 &(6a^2b - 3ab^2) + (4a^2b + 3ab^2) \\
 &= (6a^2b + 4a^2b) + (-3ab^2 + 3ab^2) \\
 &= 10a^2b + 0 \\
 &= 10a^2b
 \end{aligned}$$

Vertical Method:

$$\begin{array}{r}
 6a^2b - 3ab^2 \\
 4a^2b + 3ab^2 \\
 \hline
 10a^2b + 0 \quad \text{or } 10a^2b
 \end{array}$$

Rule for Subtracting Polynomials

To subtract a polynomial from another polynomial, change the sign of the subtrahend and proceed to addition. Keep in mind that in adding polynomials, you can only add those that are similar or like terms.

$$a - b = a + (-b)$$

↓ minuend ↓ subtrahend

Example 1. Subtract $2a + 5b$ from $3a + 8b$ **Vertical Method****Solution:**

$$\begin{array}{r}
 3a + 8b \\
 - 2a - 5b \\
 \hline
 1a + 3b \quad \text{or } a + 3b
 \end{array}$$

Horizontal Method

$$\begin{aligned}
 &(3a + 8b) - (2a + 5b) \\
 &= 3a + 8b - 2a - 5b \\
 &= 3a - 2a + 8b - 5b \\
 &= (3 - 2)a + (8 - 5)b \\
 &= 1a + 3b \\
 &= a + 3b
 \end{aligned}$$

Example 2: Subtract $2x + 3y - 4z$ from $3x - 2y + 5z$

Solution:

$$\begin{array}{r}
 3x - 2y + 5z \\
 \pm 2x \pm 3y \mp 4z \\
 \hline
 x - 5y + 9z
 \end{array}$$

Example 3: Subtract $4x - y + 2$ from the sum of $3x + 8y + 1$ and $2x + 3y - 6$

Solution: $2x + 3y - 6$

$$\begin{array}{r}
 3x + 8y + 1 \\
 \hline
 \text{Sum: } 5x + 11y - 5 \\
 \pm 4x \mp y \pm 2 \\
 \hline
 \text{Difference: } x + 12y - 7
 \end{array}$$

EXERCISE 8.1

A. Add the following:

(1) $5a, 8a$

(2) a, a

(3) $-12x, -2x$

(4) $9ab, -7ab$

(5) $6x^2 + 2x + 4, 3x^2 + 5x$

(6) $8y, 3y, -4y$

B. Find the sum of:

(1) $2a + 3b, 6a + 5b$

(2) $3x - 4y, 4x + 6y$

(3) $2a - 3b, 4a + 5b, 5a - 8b$

(4) $ab + 4cd + 6ad, 8ab + 5cd + 6ad$

(5) $4.5a + 5.7b + 3.2c, 3.5a + 1.3b + 2.8c$

(6) $1.7a + 3.2b + 4.9c, 4.3a + 3.2b + 3.1c, 6.0a + 9.6b + 3.0c$

(7) $6a + 11b + 10c, 13a + 9b + 5c, 8a + 7b + 12c$

(8) $4a + 6b + 5c, 7a + 5b + 5c, 3a + 7b + 7c$

C. Subtract the second expression from the first expression:

(1) $8a, 5a$

(2) $-7ab, 12ab$

(3) $4a, -8a$

(4) $6a + 7b, 2a + 3b$

(5) $9a + 11b + 15c, 5a + 3b - 6c$

(6) $9c + 8d + 6e, 3e - 5c + 4d$

(7) $18x + 16y + 12z, 10x + 9y + 8z$

(8) $14a + 15b + 16c, 8a + 10b - 2c$

(9) $3x^2 + 25ab + 21bc, 4x^2 + 18ab - 6bc$

Multiplying Polynomials

Unlike the processes of addition and subtraction, multiplication of polynomials considers not only the numerical coefficients but also the literal coefficients.

Rules in Multiplying Polynomials:**A. Monomial by Monomial:**

To multiply a monomial with another monomial, simply multiply

the numerical coefficients then multiply the literal coefficients by applying the basic laws of exponents.

Examples:

1. $(x^3)(x^5) = x^{3+5}$

$$= x^8$$

Applying the law of exponents, copy the common base "**x**" and add the exponents "**3**" and "**5**" Simplify the result.

2. $(3x^2)(-5x^{10}) = (3)(-5)(x^{2+10})$

$$= -15x^{12}$$

Multiply the numerical coefficients "**3**" and "**-5**" and copy the common base "**x**" then add the exponents "**2**" and "**10**". Simplify the result.

B. Monomial by a Polynomial:

To multiply a monomial with a polynomial, simply apply the distributive property and follow the rule in multiplying monomial by a polynomial. Multiplication of monomial by a polynomial could also be done vertically. In the examples below, the solutions are presented in two ways:

Examples:

i) $(3x)(x^2 - 5x + 7)$

Solutions:

Using the distributive property,

$$\begin{aligned} & (3x)(x^2 - 5x + 7) \\ &= (3x)(x^2) + (3x)(-5x) + (3x)(7) \\ &= (3x^{1+2}) + (-15x^{1+1}) + (21x) \\ &= 3x^3 - 15x^2 + 21x \end{aligned}$$

Using the Vertical Solving,

$$\begin{array}{r} x^2 - 5x + 7 \\ \times \quad 3x \\ \hline 3x^3 - 15x^2 + 21x \end{array}$$

Examples:

ii) $(-5x^2y^3)(2x^2y - 3x + 4y^5)$

Solutions:

Using the distributive property,

$$\begin{aligned} & (-5x^2y^3)(2x^2y - 3x + 4y^5) \\ &= (-5x^2y^3)(2x^2y) + (-5x^2y^3)(-3x) + (-5x^2y^3)(4y^5) \\ &= (-5)(2)(x^{2+2})(y^{3+1}) + (-5)(-3)(x^{2+1})(y^3) + (-5)(4)(x^2)(y^{3+5}) \\ &= -10x^4y^4 + 15x^3y^3 - 20x^2y^8 \end{aligned}$$

Using the Vertical Solving,

$$\begin{array}{r} 2x^2y - 3x + 4y^5 \\ \times \quad -5x^2y^3 \\ \hline -10x^4y^4 + 15x^3y^3 - 20x^2y^8 \end{array}$$

C. Binomial by a Binomial

To multiply a binomial with another binomial, you can use the F-O-I-L Method or Smile Method. F-O-I-L stands for F-First, O-Outer, I-Inner, and L-Last. This acronym represents the order in which you should multiply the binomial terms to get the product. You can draw arcs from the first to first terms and outer to outer terms over the top of the expression and draw arcs from the inner to inner terms and last to last terms on the bottom. When you do, you create a smiley face!

Another way is the vertical way of multiplying, which is the conventional one.

Examples:

i) $(x + 3)(x + 5)$

Solutions:

Using F-O-I-L Method:

$$\begin{array}{ccc} \text{First terms} & & \text{Last terms} \\ & \nearrow & \nwarrow \\ (x + 3)(x + 5) & & \\ & \nwarrow & \nearrow \\ & \text{Inner terms} & \\ & \nwarrow & \nearrow \\ & \text{Last terms} & \end{array}$$

$$\begin{aligned} \text{First terms: } & \longrightarrow (x)(x) = x^2 \\ \text{Outer terms: } & \longrightarrow (x)(5) = 5x \\ \text{Inner terms: } & \longrightarrow (3)(x) = 3x \\ \text{Last terms: } & \longrightarrow (3)(5) = 15 \end{aligned}$$

Since $5x$ and $3x$ are similar terms, we can combine them: $\longrightarrow 5x + 3x = 8x$.

The final answer is $x^2 + 8x + 15$.

Using the Vertical Solving,

$$\begin{array}{r}
 \begin{array}{c}
 x + 3 \\
 \nearrow \quad \nwarrow \\
 \uparrow \quad \downarrow \\
 x \quad x + 5 \\
 \hline
 x^2 + 3x \\
 + \quad 5x + 15 \\
 \hline
 x^2 + 8x + 15
 \end{array}
 \end{array}
 \begin{array}{l}
 \longrightarrow \text{Multiply } x \text{ to } (x + 3) \\
 \longrightarrow \text{Multiply } 5 \text{ to } (x + 3), \text{ Align similar terms}
 \end{array}$$

Notice that any of the two solution will give you the same answer which is $x^2 + 8x + 15$.

D. Polynomial with More Than One Term to Polynomial with Three or More Terms

To multiply a polynomial with more than one term to a polynomial with three or more terms, simply apply the Distributive Property of Multiplication over Addition. Multiply the first term of the first polynomial to each term of the other polynomial. Repeat the procedure up to the last term and simplify the results by combining similar terms.

Another way of multiplying this is using the vertical way, which is the conventional one.

Example:

1. $(x + 3)(x^2 - 2x + 3)$

Solution:

Using the distributive property:

$$\begin{aligned}
 &(x + 3)(x^2 - 2x + 3) \\
 &= x(x^2 - 2x + 3) + 3(x^2 - 2x + 3) \\
 &= (x^3 - 2x^2 + 3x) + (3x^2 - 6x + 9) \\
 &= x^3 + x^2 - 3x + 9
 \end{aligned}$$

Using the Vertical Solving,

$$\begin{array}{r}
 \begin{array}{c}
 x^2 - 2x + 3 \\
 \nearrow \quad \nwarrow \quad \nwarrow \\
 \uparrow \quad \downarrow \quad \downarrow \\
 x \quad x + 3 \\
 \hline
 x^3 - 2x^2 + 3x \\
 + \quad 3x^2 - 6x + 9 \\
 \hline
 x^3 + x^2 - 3x + 9
 \end{array}
 \end{array}$$

Dividing Polynomials

Like in multiplication, the division of polynomials applies the laws of exponents.

Rules in Dividing Polynomials

A. Polynomial by Monomial

To divide a polynomial by a monomial, simply divide each term of the polynomial by the given divisor.

Examples:

1. Divide $12x^4 - 16x^3 + 8x^2$ by $4x^2$.

Solution:

$$\begin{aligned}\frac{12x^4 - 16x^3 + 8x^2}{4x^2} &= \frac{12x^4}{4x^2} - \frac{16x^3}{4x^2} + \frac{8x^2}{4x^2} \\ &= 3x^{4-2} - 4x^{3-2} + 2x^{2-2} \\ &= 3x^2 - 4x + 2x^0 \\ &= 3x^2 - 4x + 2(1) \\ &= 3x^2 - 4x + 2\end{aligned}$$

Examples:

1. Divide $x^2 - 3x + 10$ by $x + 2$

Solution:

$x^2 - 3x + 10$ and $x + 2$ are already in standard form.

$\begin{array}{r} x - 5 \\ x + 2 \overline{) x^2 - 3x + 10} \\ \underline{x^2 - 2x} \\ -5x - 10 \\ \underline{+5x + 10} \\ 0 \end{array}$	→ Divide x^2 by x and write the result on top. → Multiply the result x to $x + 2$. → Subtract the product $x^2 + 2x$ (subtrahend) from $x^2 - 3x$ (minuend). → Remember to change the sign of the subtrahend then proceed to addition. Bring down the remaining term -10. Then, divide $-5x$ by x. → Multiply the result -5 to $x + 2$. → Subtract the product $-5x - 10$ from $-5x - 10$. Thus, the remainder is 0.
--	---

Therefore, the quotient is $x - 5$.

2. Divide $x^3 - 6x^2 + 11x + 6$ by $x - 3$

Solution:

Since the terms in the dividend and divisor are in standard form, then,

$$\begin{array}{r}
 x^2 - 3x + 2 \\
 x - 3 \overline{) x^3 - 6x^2 + 11x - 6} \\
 \underline{x^3 - 3x^2} \\
 -3x^2 + 11x \\
 \underline{+3x^2 - 9x} \\
 2x - 6 \\
 \underline{2x - 6} \\
 0
 \end{array}$$

- Divide x^3 by x and write the result on top.
- Multiply the result x^2 to $x - 3$. Subtract the product $x^3 - 3x^2$ (subtrahend) from $x^3 - 6x^2$ (minuend).
- Remember to change the sign of the subtrahend then proceed to addition. Bring down $11x$. Then, divide $-3x^2$ by x .
- Multiply the result $-3x$ to $x - 3$.
- Subtract the product $-3x^2 + 9x$ from $-3x^2 + 11x$ and bring down -6 . Then, divide $2x$ by x .
- Multiply the result 2 to $x - 3$.
- Subtract the product $2x - 6$ from $2x - 6$. Thus, the remainder is 0 .

Therefore, the quotient is $x^2 - 3x + 2$.

EXERCISE 8.2

I. Find the product of the following expressions: (Monomial with monomial)

- (1) $15a, 8a$ (2) $9a, 6b$ (3) $10a^2, 2a^2$ (4) $4e^2, 8cd^2$
 (5) $6a^2b, 5b^3$ (6) $-3a, 2b, -3$ (7) $4m^2, -5m^2, 3m^3$
 (8) $-6xy, 3x^2y^2, 2xy$

II. Find the product of the following expressions: (Monomial with binomial)

- (1) $a + b, 9$ (2) $x - y, -4y$ (3) $2m^2 - 3n^2, -5mn$
 (4) $3xy + 2x^2y, -6xy^2$ (5) $6lm - 4mn, 3t^2m^2n$
 (6) $-5bc + 3cad, -2bc^2a$

III. Find the product of the following expressions: (Binomial with binomial)

- (1) $8c + 12d, -3c^2 + 2cd$ (2) $a + b, a - b$ (3) $a^2 + b^2, a^2 - b^2$

(4) $3x - 2y, 2x + 5y$ (5) $2lm + 3mn, -2m + 3n$

(6) $8pq^3 - 3p^2q^2, 2pq + 2p$

IV. Find the product of the following expressions: (Monomial with trinomial)

(1) $2a - 3b + 4c, 2abc$

(2) $a^2 + b^2 - c^2, ab$

(3) $m^2n^2 + mn^2 - mn, 2mn$

(4) $x^2 - 6xy + 9y^2, -3xy$

(5) $2p^2 - 3pq + q^2, 4pq^2$

(6) $xyz + x^2y^2 - yz^2, xyz$

V. Find the product of the following expressions: (Binomial with trinomial)

(1) $1 + 2x - 3x^2, 1 - x$

(2) $8c + 12d - 6e, c^2 - d^2$

(3) $a^2 - ab - b^2, a - b$

(4) $x^2 - 5x + 4, 3x - 4$

(5) $a^2 - 2a^2b^2 + b^2, a^2 - b^2$

(6) $3pq^2 + 2pq^3 - q^2, p^2 - q^2$

VI. Perform the division.

(1) $(x^3 - 1) \div (x - 1)$

(2) $(x^2 - 5x + 6) \div (x - 2)$

(3) $(8x^3 + y^3) \div (2x + y)$

(4) $(x^4 - y^4) \div (x - y)$

(5) $(8a^3 - 7a - 1) \div (a - 1)$